

Discrete Mathematics

Core Concepts

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Discrete Mathematics

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Sequences

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Graph Theory

Paths & Connectivity

What is *Discrete Mathematics*?



Core Definition

Discrete Mathematics is the study of **countable, distinct structures**, encompassing areas such as logic, sets, integers, and graphs, rather than continuous quantities.

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Computer Science

Foundations of computing and formal reasoning



Algorithms

Design and analysis of efficient algorithms



Cryptography

Security and coding theory applications



Logical Thinking

Develops precise reasoning and proof skills

Propositions and Logical Operators

Propositions

A proposition is a declarative statement that is either true or false. These are the fundamental building blocks of logical reasoning.

Examples

TRUE p : "2 is even"

FALSE q : "The sky is green"

Compound Propositions: Built from atomic propositions using logical operators

Logical Operators

NOT ($\neg p$): Negation

Changes the truth value. If p is true, $\neg p$ is false.

AND ($p \wedge q$): Conjunction

True only if both p and q are true.

OR ($p \vee q$): Inclusive Disjunction

True if p is true, or q is true, or both are true.

IMPLIES ($p \rightarrow q$): Implication

"If p , then q ." False only if p is true and q is false.

IFF ($p \leftrightarrow q$): Biconditional

" p if and only if q ." True if p and q have the same truth value.

Truth Tables and Logical Equivalences

Truth Table for Logical Operators

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

Note: Truth tables systematically show all possible truth value combinations

Key Logical Equivalences

De Morgan's Laws

$$\neg(p \wedge q) \equiv (\neg p \vee \neg q)$$

$$\neg(p \vee q) \equiv (\neg p \wedge \neg q)$$

Implication Equivalence

$$p \rightarrow q \equiv (\neg p \vee q)$$

Contrapositive

$$(p \rightarrow q) \equiv (\neg q \rightarrow \neg p)$$

Biconditional

Inference Rules and Proof Techniques

Fundamental Inference Rules

Modus Ponens (MP)

$$p, (p \rightarrow q) \Rightarrow q$$

If it rains, the ground is wet. It rains. Therefore, the ground is wet.

Modus Tollens (MT)

$$\neg q, (p \rightarrow q) \Rightarrow \neg p$$

If n is even, n^2 is even. n^2 is not even. Therefore, n is not even.

Hypothetical Syllogism (HS)

$$(p \rightarrow q), (q \rightarrow r) \Rightarrow (p \rightarrow r)$$

Chains implications together to form a new implication.

Disjunctive Syllogism (DS)

$$(p \vee q), \neg p \Rightarrow q$$

Proof Techniques



Direct Proof

Assumes the premise is true and logically deduces the conclusion step by step.



Proof by Contrapositive

To prove $p \rightarrow q$, prove its contrapositive: $\neg q \rightarrow \neg p$. Both statements are logically equivalent.



Proof by Contradiction

Assumes the negation of what you want to prove and derives a contradiction, proving the original statement must be true.



Mathematical Induction

Proves statements about natural numbers using base case and inductive step: if $P(k)$ is true, then $P(k+1)$ is true.

Mathematical Induction Example



Claim:

The sum of the first n natural numbers is given by:

$$1 + 2 + \dots + n = n(n+1)/2$$

1

Base Case

$$n = 1$$

$$\text{LHS} = 1$$

$$\text{RHS} = 1(1+1)/2 = 1$$

✓ LHS = RHS
Formula holds for $n = 1$

2

Inductive Hypothesis

Assume the formula holds for some arbitrary integer $k \geq 1$.

$$1 + 2 + \dots + k = k(k+1)/2$$

That is, we assume this equation is true for $n = k$

3

Inductive Step

Show the formula holds for $k+1$:

Sum of first $(k+1)$ numbers:

$$\begin{aligned} 1 + 2 + \dots + k + (k+1) \\ &= k(k+1)/2 + (k+1) \\ &= (k+1)(k/2 + 1) \\ &= (k+1)(k+2)/2 \quad \checkmark \end{aligned}$$

✓ Formula verified for $k+1$

Set Theory Fundamentals

What is a Set?

A well-defined collection of distinct objects, called **elements**.

Set Representation



Roster Notation

Lists all elements explicitly.

$$A = \{1, 3, 5\}$$



Set-builder Notation

Describes elements by a property.

$$: x \text{ is odd, } x \leq 5$$

Key Concepts



Element

Object belonging to a set. $a \in A$



Empty Set

Set with no elements. $\emptyset$



Subset

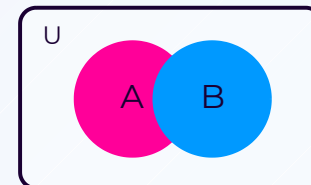
$A \subseteq B$ if every element of A is in B



Cardinality

Number of elements. $|A|$

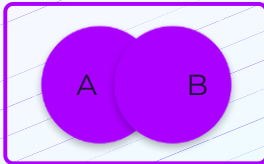
Venn Diagram



Set Operations

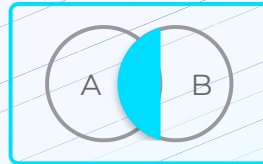
$A \cup B$

All elements in A or B



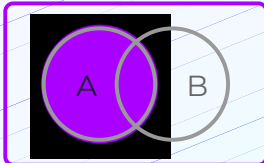
$A \cap B$

Elements common to A and B



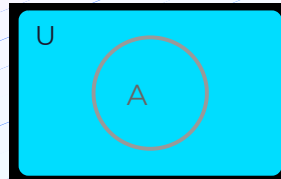
Difference $(A - B)$

Elements in A but not in B



Complement (A')

Elements in U but not in A



De Morgan's Laws

Fundamental laws relating complement, union, and intersection operations.

$(A \cup B)' = A' \cap B'$

The complement of a union equals the intersection of complements

$(A \cap B)' = A' \cup B'$

The complement of an intersection equals the union of complements



Key Insight: De Morgan's Laws show duality

Set Operations: *Union*

$A \cup B$: All elements in A or B

Definition

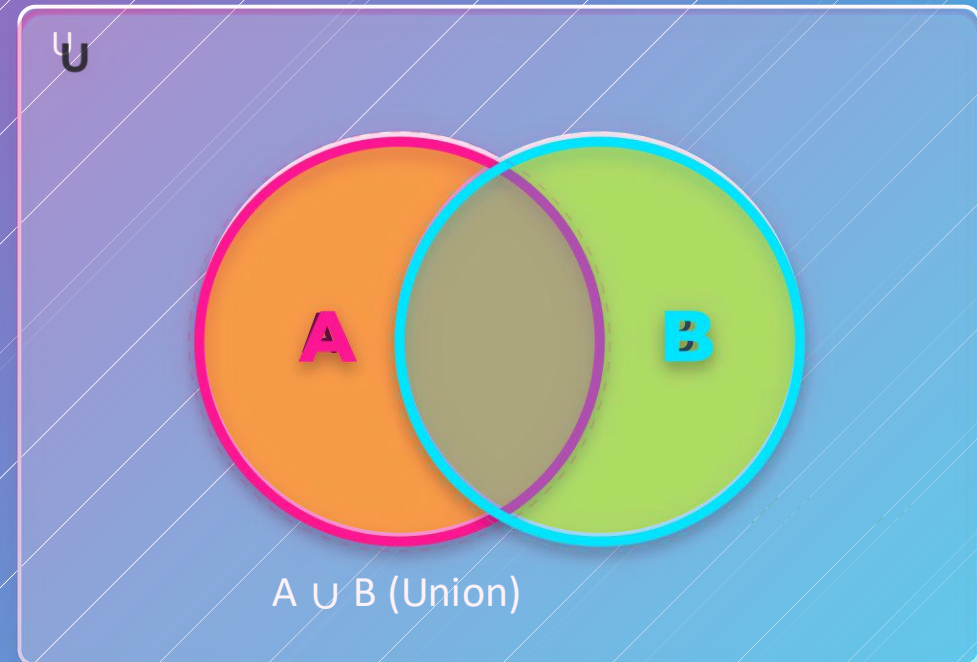
The union of sets A and B contains all elements that belong to A, or to B, or to both.

Example

$A = \{1, 2, 3\}$

$B = \{3, 4, 5\}$

$A \cup B = \{1, 2, 3, 4, 5\}$



Relations and Their Properties

Binary Relations

A binary relation R between sets A and B is any subset of the Cartesian product $A \times B$. If $A = B$, it is a relation on A .

Notation:

$(a, b) \in R$ means "a is related to b"

Relations describe connections between elements of sets

Key Properties of Relations

Reflexive

For all $a \in A$, $(a, a) \in R$

Symmetric

If $(a, b) \in R$, then $(b, a) \in R$

Antisymmetric

If $(a, b) \in R$ and $(b, a) \in R$, then $a = b$

Transitive

If $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$

Equivalence Relations

A relation that is reflexive, symmetric, and transitive. Equivalence relations partition a set into disjoint equivalence classes.

Functions and Mappings



Function Definition

A function $f: A \rightarrow B$ is a relation that assigns each element $a \in A$ to exactly one element $b \in B$.

Key Components



Domain (A)

The set of all allowed input values



Codomain (B)

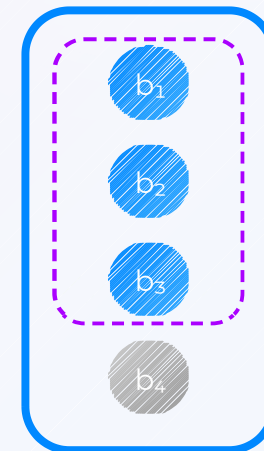
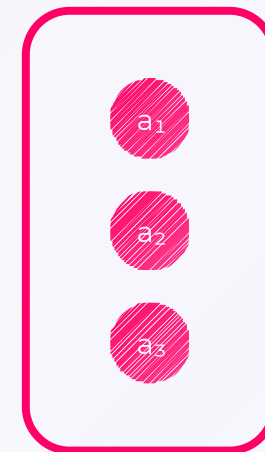
The set containing all possible outputs



Range / Image ($f(A)$)

The set of all actual output values

Visual Representation



Codomain (B)

Types of Functions

Injective (One-to-one)

Distinct inputs map to distinct outputs. If $f(a) = f(b)$, then $a = b$.

✓
Example: $f(x) = 2x$ on integers

✗
Counterexample: $f(x) = x^2$ on integers

Surjective (Onto)

Every element in the codomain has at least one corresponding element in the domain.

✓
Example: $f(x) = x^3$ from $\mathbb{R}$ to $\mathbb{R}$

✗
Counterexample: $f(x) = x^2$ from $\mathbb{R}$ to $\mathbb{R}$

Bijjective (One-to-one and Onto)

A function that is both injective and surjective. Perfect pairing between domain and codomain.

Inverse Functions

Existence Condition

A function f has an inverse f^{-1} if and only if f is bijective.

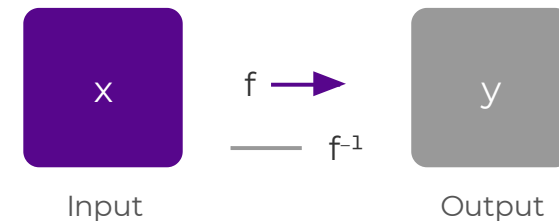
Properties

The inverse function "undoes" the original function:

1 $f(f^{-1}(x)) = x$

2 $f^{-1}(f(y)) = y$

Visual Example



Sequences and Patterns

Arithmetic Sequence

Definition

Each term is obtained by adding a constant difference to the preceding term.

Formula

$$a_n = a_1 + (n - 1) \times d$$

where a_1 is the first term, d is the common difference

Example

If $a_1 = 2$ and $d = 3$, the sequence is:

2 +3 5 +3 8 +3 11 ...

Geometric Sequence

Definition

Each term is obtained by multiplying the preceding term by a constant ratio.

Formula

$$a_n = a_1 \times r^{(n-1)}$$

where a_1 is the first term, r is the common ratio

Example

If $a_1 = 3$ and $r = 2$, the sequence is:

3 ×2 6 ×2 12 ×2 24 ...

Fibonacci Sequence

Visual Representation

Recurrence Relation:

$$F_n = F_{n-1} + F_{n-2}$$

Initial Conditions: $F_0 = 0, F_1 = 1$

0

F_0

First Term

1

F_1

Second Term

1

F_2

$0 + 1 = 1$

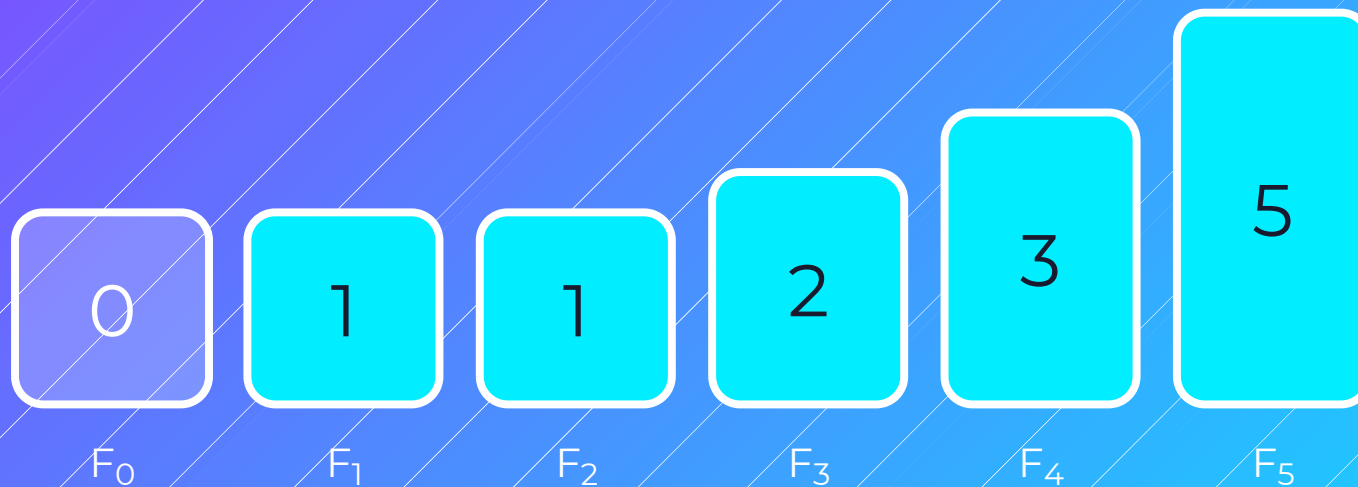
Fibonacci Sequence

Visual Representation

Recurrence Relation:

$$F_n = F_{n-1} + F_{n-2}$$

Initial Conditions: $F_0 = 0, F_1 = 1$



Recurrence Relations

Definition

A recurrence relation defines each term of a sequence based on one or more preceding terms. To fully define a sequence using a recurrence relation, initial conditions (base cases) are also required.

Key Components

- Initial Conditions: Starting values
- Recurrence Rule: Formula relating terms
- Sequence: Generated terms

Fibonacci Sequence

Initial Conditions

$$F_0 = 0, F_1 = 1$$

Recurrence Relation

$$F_n = F_{n-1} + F_{n-2}, \text{ for } n \geq 2$$

Each Fibonacci number is the sum of the two preceding ones.

First Few Terms



Counting Principles

Permutations

Order Matters

Formula

$${}^n P_r = n! / (n - r)!$$

Example

Arrange 3 of 5 letters

$${}^5 P_3 = 5 \times 4 \times 3 = 60 \text{ arrangements}$$

When to Use

- Rankings/arrangements
- Passwords without repetition
- Seating arrangements

Combinations

Order Doesn't Matter

Formula

$${}^n C_r = n! / (r!(n - r)!)$$

Example

Choose a 3-person committee from 5

$${}^5 C_3 = 10 \text{ committees}$$

When to Use

- Forming committees
- Lottery picks
- Selecting a subset

Graph Theory Basics

Definition

A graph G is an ordered pair (V, E) , where:

V is a set of vertices (nodes)

E is a set of edges connecting pairs of vertices

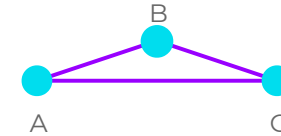
Key Concepts

-  **Vertex (Node)**
Fundamental unit representing an entity
-  **Edge (Link)**
Connection between two vertices
-  **Degree of Vertex**
Number of edges incident to a vertex

Types of Graphs

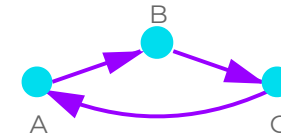
Undirected Graph

Edges have no orientation; connections are bidirectional.



Directed Graph (Digraph)

Edges have specific orientation, represented as ordered pairs.



Simple Graph

No loops or multiple edges between the same pair of vertices.

Paths, Cycles, and Connectivity

Core Definitions

W

Walk

A sequence of vertices and edges, where repetitions are allowed.

T

Trail

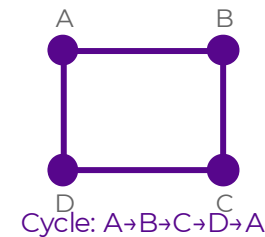
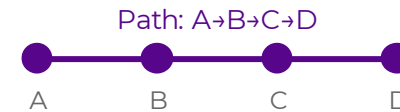
A walk with no repeated edges.

P

Path

A walk with no repeated vertices (and thus no repeated edges).

Visual Examples

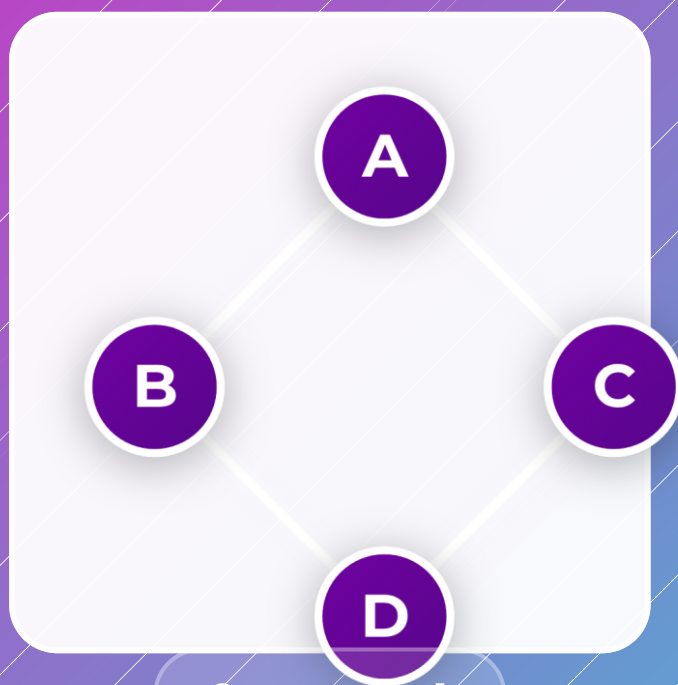


Connectivity

A graph is **connected** if there exists a path between every pair of vertices. Otherwise, it is **disconnected**.

Graph Connectivity: Disconnected

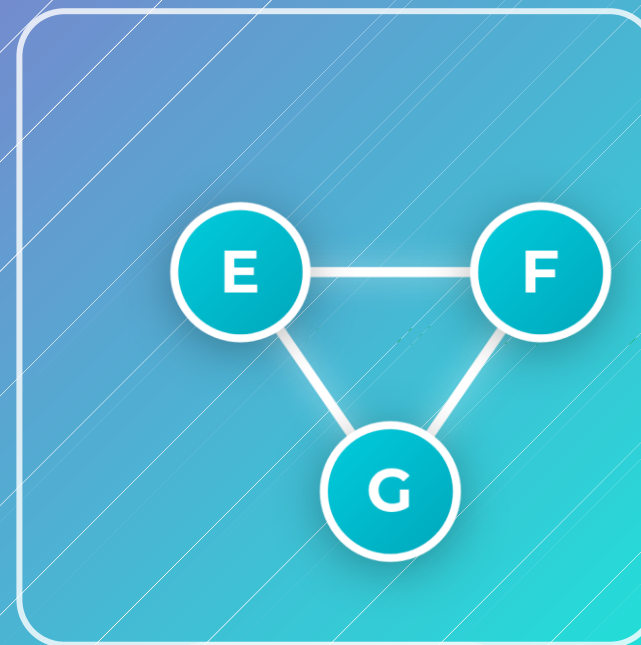
A graph is **disconnected** if there exist vertices with no path between them



Component 1



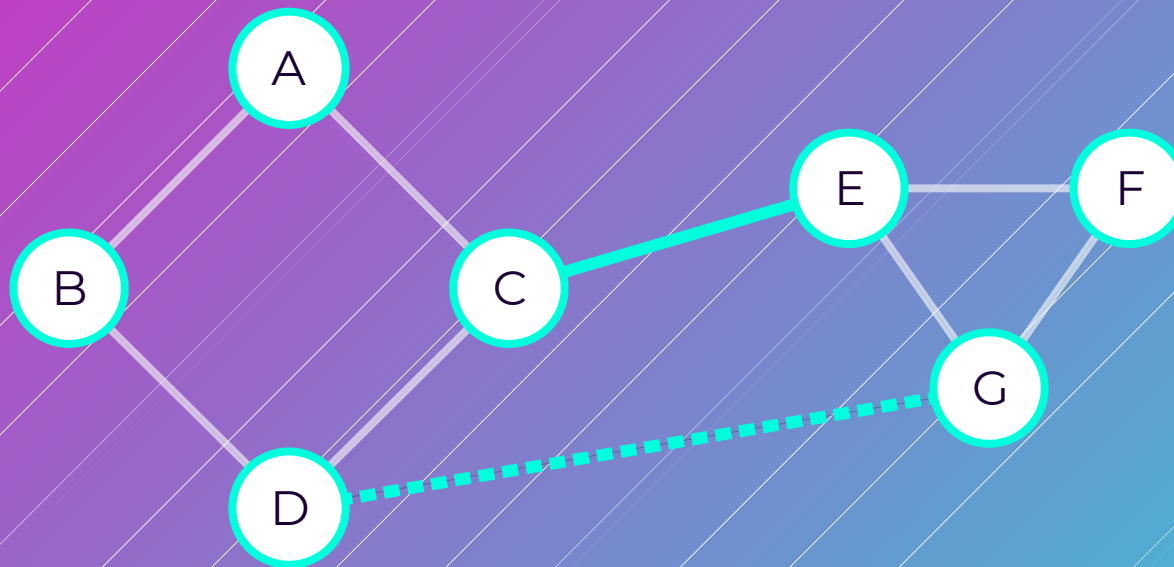
No Path



Component 2

Graph Connectivity

A graph is **connected** if there exists a path between every pair of vertices



✓
Paths Exist

Connected Graph

Example path from **A** to **G**: $A \rightarrow C \rightarrow E \rightarrow G$

Graph Theory Applications

Powerful tools for modeling and solving problems in diverse fields



Networks & Routing

Models communication networks, transportation systems, and computer networks, enabling efficient data flow and route optimization.

- Real-time optimization



Shortest Paths (Dijkstra's)

Algorithms like Dijkstra's find minimum-cost routes between points in weighted graphs, critical for navigation and logistics.

- GPS & Navigation



Scheduling (DAGs)

Directed Acyclic Graphs represent tasks with dependencies, allowing topological sorting to determine valid execution order.

- Project management



Social Graphs

Model relationships between individuals in social networks, facilitating community detection, influence analysis, and information spread.

- Social media analytics

Summary and Key Takeaways

8 Core Topics Mastered

- | | |
|---|---|
| 1 Logic
Propositions, operators, truth tables | 2 Set Theory
Operations, identities, notation |
| 3 Relations
Properties, equivalence classes | 4 Functions
Domain, codomain, types |
| 5 Proofs
Direct, contradiction, induction | 6 Sequences
Arithmetic, geometric, recurrence |
| 7 Counting
Permutations, combinations | 8 Graph Theory
Paths, cycles, connectivity |

Key Takeaways

- Utilize logical equivalences and truth tables for rigorous reasoning
- Model and analyze data using sets, relations, and functions
- Apply proof techniques, especially mathematical induction
- Count outcomes efficiently and compute probabilities

What's Next

- P Practice:** Solve problem sets to solidify understanding
- E Explore:** Advanced topics in number theory
- A Apply:** Use concepts in real-world algorithms



Thank You

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Questions?

NBU Computer Science Department

Discrete Mathematics Core Concepts